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Summary of my Bachelor Thesis: Environmental Kuznets Curve & Spatial Econometrics

1. Project Objective and Implementation

My thesis investigates the Environmental Kuznets Curve (EKC) across 213 European NUTS 2 regions (2014–2022), replicating and extending David Maddison's (2006) spatial econometric framework. The empirical execution employs a First-Difference ($\Delta \ln$) panel estimator and econometric modelling, integrated then with spatial weight matrices (W) to model transboundary spillovers, moving to areal data modelling and spatial econometrics. We process and harmonize multi-source data from EDGAR v8.0 and Eurostat to evaluate whether regional economic growth decouples from emissions (CO_2, CH_4, N_2O, F-gases).

2. Data Wrangling

- **EDGAR v8.0 (EC-JRC):** NUTS2 emission inventories (*ktons*) across all economic sectors for Fossil CO_2, CH_4, N_2O, and F-gases (GWP_{100} AR5).
- **Eurostat NAMA & DEMO:** Regional GDP and total resident population.
- **Eurostat GISCO:** EPSG:3035.

Made dataset tidy, found the per-capita value, performed first difference and joint the dataset.

Final Balanced Dataset Sample ($N = 213$ Regions, 852 Observations)

NUTS ID	Subst	E_{2014}	E_{2022}	GDP_{2014}	GDP_{2022}	$\Delta \ln(E)$	$\Delta \ln(\text{GDP})$	$\Delta(\ln(\text{GDP})^2)$
AT11	CO_2	6.593×10^{-3}	6.034×10^{-3}	26,608	34,771	−0.0886	+0.2676	+5.5240
AT11	CH_4	6.257×10^{-4}	6.087×10^{-4}	26,608	34,771	−0.0275	+0.2676	+5.5240
AT11	F-gases	2.562×10^{-4}	2.982×10^{-4}	26,608	34,771	+0.1520	+0.2676	+5.5240
AT11	N_2O	7.504×10^{-4}	6.677×10^{-4}	26,608	34,771	−0.1168	+0.2676	+5.5240
AT12	CO_2	1.030×10^{-2}	8.764×10^{-3}	32,016	41,036	−0.1616	+0.2482	+5.2119
AT12	CH_4	1.371×10^{-3}	1.238×10^{-3}	32,016	41,036	−0.1020	+0.2482	+5.2119
AT12	F-gases	3.023×10^{-4}	3.439×10^{-4}	32,016	41,036	+0.1290	+0.2482	+5.2119
AT12	N_2O	6.734×10^{-4}	5.940×10^{-4}	32,016	41,036	−0.1254	+0.2482	+5.2119

3. Econometric Modeling & Diagnostic Testing

Estimation of the model via First Differences using `tidyverse`, `lmtest`, and `tseries`:

$$\Delta \ln(E_i) = \beta_1 \Delta \ln(\text{GDP}_i) + \beta_2 \Delta(\ln(\text{GDP}_i)^2) + \Delta\varepsilon_i$$

I implemented the OLS estimation across pollutants and evaluated residual validity against Gauss-Markov assumptions via Jarque-Bera normality tests and White heteroskedasticity tests:

```
# Libraries: tidyverse, lmtest, tseries
# OLS estimation in first differences & specification tests per substance
model   <- lm(diff_ln_E ~ diff_ln_GDP + diff_ln_GDP_sq, data = data_sub)
jb_p    <- jarque.bera.test(residuals(model))$p.value
white_p <- bptest(model, ~ diff_ln_GDP * diff_ln_GDP_sq +
              I(diff_ln_GDP^2) + I(diff_ln_GDP_sq^2), data = data_sub)$p.value
```

While the inverted-U is confirmed ($\beta_1 > 0, \beta_2 < 0$), all specifications reject error normality ($p < 0.05$), indicating structural omission of dependence.

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4. Spatial Weight & Geometric Modeling

Using `sf` and `spdep` package, I mapped the spatial structure by calculating geometric centroids from official NUTS 2 polygons and constructed three row-standardized spatial weight matrices W ($w^*_{ij} = w_{ij} / \sum_j w_{ij}$): 1. **Inverse Distance with Critical Threshold** (378 km = 1 -NN max distance): 2. **Inverse Distance Unlimited:** Dense connectivity modeling long-range decay: $w_{ij} = 1/d_{ij}$. 3. **k-Nearest Neighbors** ($k = 5$): Uniform spatial connectivity ($w^*_{ij} = 0.20$).

```
# Libraries: sf, spdep
# 1st Weigth matrix
coords <- st_coordinates(st_centroid(st_geometry(sf_all %>% filter(Substance == "CO2")),
                        of_largest_polygon = TRUE))
max_1nn <- max(unlist(nbdists(knn2nb(knearneigh(coords, k = 1)), coords))) #k-NN threshold
nb_threshold <- dnearneigh(coords, d1 = 0, d2 = max_1nn)
idw_threshold <- lapply(nbdists(nb_threshold, coords), function(x) 1/x)
W_inv_dist_threshold <- nb2listw(nb_threshold,
                        glist = idw_threshold, style = "W", zero.policy = TRUE)
```

5. Spatial test and estimation

Test for spatial autocorrelation and select the appropriate model (SAR vs. SEM). First I performed Moran's I on OLS residuals and then Lagrange Multiplier tests (`lm.LMtests`, including Robust LM). Models were then estimated via Maximum Likelihood and FGLS using `spatialreg`:

```
# Libraries: spdep, spatialreg
# 1. Spatial diagnostics on regression residuals
morans_lm <- lm.morantest(ekc_model, W_knn5_rowstd)
morans_lm <- lm.morantest(ekc_model, W_inv_dist_unlimited)
morans_lm <- lm.morantest(ekc_model, W_inv_dist_threshold)
LM_sem    <- lm.LMtests(ekc_model, W_knn5_rowstd, test = c("LMerr", "RLMerr"))
```

```
LM_sar    <- lm.LMtests(ekc_model, W_inv_dist_unlimited, test = c("LMlag", "RLMlag"))

# 2. Maximum Likelihood / FGLS spatial estimation
sem_Fgases <- errorsarlm(diff_ln_E ~ diff_ln_GDP + diff_ln_GDP_sq,
                                 data = df_Fgases, listw = W_knn5_rowstd)
sar_N2O    <- lagsarlm(diff_ln_E ~ diff_ln_GDP + diff_ln_GDP_sq,
                                 data = df_N2O, listw = W_inv_dist_unlimited)
```

Comparative Estimation Results

Model / Substance	$\Delta \ln(\text{GDP})$ [p-val]	$\Delta(\ln \text{GDP})^2$ [p-val]	Spatial Parameter [p-val]
OLS: CO_2	$+3.170$ [< 0.001]	-0.153 [< 0.001]	—
OLS: CH_4	$+2.105$ [< 0.001]	-0.092 [0.003]	—
SAR: N_2O *(Unlimited W)*	$+2.132$ [< 0.001]	-0.100 [< 0.001]	$\boldsymbol{\rho = 0.931}$ [< 0.001]
SEM: F-gases *(k-NN 5)*	$+2.271$ [0.052]	-0.058 [0.339]	$\boldsymbol{\lambda = 0.865}$ [< 0.001]